

Student Competitions That Produce Real Work

A BlueRobins Field Guide for the 2026–27 Season

Every competition in this guide meets one standard: a student cannot win it in an afternoon. Each one requires a project built over weeks or months, guided by a mentor, that ends in something real — a research paper, a working prototype, an app, a documentary, a business plan, a portfolio, an experiment that flies.

That standard is deliberate. Timed tests and one-sitting contests measure what a student already knows. Project competitions build what a student can do. The difference shows up in the work itself, and it is the difference admissions readers, scholarship committees and research supervisors have learned to look for.

39 competitions across 12 sectors, from grade 3 through grade 12.

How to read this guide

Competitions are grouped by grade band, because eligibility is the first filter any family applies. Within each band, entries are ordered roughly by the depth of work they demand.

Every entry answers the same eight questions:

- **Sector** — the field the work sits in, for matching a student to the right arena
- **Who it's for** — exact grade or age eligibility
- **What students produce** — the artifact that comes out the other end
- **What it builds** — the capability the student carries forward
- **Recognition** — selectivity, awards and what winning actually confers
- **Key dates** — the 2026–27 season
- **Investment** — full cost, including the parts that arrive later
- **Where a mentor fits** — how outside guidance is treated by the rules

Sectors at a glance

Sector	Competitions in this guide
Scientific Research	Regeneron ISEF · Regeneron Science Talent Search · JSHS · Thermo Fisher JIC · MIT THINK · Davidson Fellows
Invention & Product Design	Invention Convention · Lemelson-MIT InvenTeams · 3M Young Scientist Challenge
Robotics & Engineering	FIRST Robotics & FIRST Tech Challenge · VEX IQ · FIRST LEGO League · Future City · Zero Robotics
Aerospace & Space Science	NASA TechRise · Genes in Space · Zero Robotics
Computer Science & AI	Congressional App Challenge · Technovation Girls
Entrepreneurship & Business	Diamond Challenge · Blue Ocean · Conrad Challenge · NFTE World Series of Innovation
Environment & Sustainability	Stockholm Junior Water Prize · Bow Seat Ocean Awareness · GENIUS Olympiad
Agriculture & Food Security	World Food Prize Youth Institutes

Sector	Competitions in this guide
Data Science & Quantitative Analysis	Modeling the Future Challenge
Writing & Humanities	The Concord Review · JFK Profile in Courage · National History Day · Scholastic Art & Writing
Film, Media & Communication	C-SPAN StudentCam · All American High School Film Festival · Breakthrough Junior Challenge
Arts & Design	YoungArts · Scholastic Art & Writing
Global Leadership & Social Impact	Rise · The Junior Academy · World Food Prize · NFTE

A note on the younger bands

Nothing a third- or seventh-grader enters appears on a college application, and families should never be told otherwise. What the elementary and middle school competitions do is build the specific capability the high school competitions reward — the habit of keeping a design logbook, the discipline of an abstract, the experience of defending your work to a judge who asks hard questions.

That is why every entry for grades 3–8 in this guide names **where it leads**: the specific 9–12 program, usually run by the same organization, that the younger competition feeds. A student who does FIRST LEGO League in fourth grade and FIRST Tech Challenge in eighth is not collecting credentials. They are eight years into a discipline by the time it counts.

What this guide excludes, and why

Spelling bees, quiz bowl, math league, geography bee and timed subject olympiads are excellent activities and appear nowhere in this guide. They produce no artifact, and there is no project for a mentor to guide. A student's performance in them is essentially fixed before the day begins.

Also excluded: any award where every entrant receives a certificate, where a nomination carries a fee, or where publication is contingent on payment. These are widely marketed to families and widely discounted by the people they are meant to impress.

Every competition that follows survives both filters.

GRADES 3–5

1. Invention Convention Worldwide

The country's largest invention-education pipeline, and the only one that runs unbroken from kindergarten through grade 12. Students identify a problem they have personally encountered, invent something that solves it, and defend the invention to judges who score the *process* more heavily than the product.

Sector	Invention & Product Design
Who it's for	Grades K–12. Teams up to four; a team competes in the grade band of its oldest member
What students produce	A working invention or prototype, an inventor's logbook documenting the whole design process, a display board, and a recorded presentation

What it builds	Structured invention methodology — problem identification, prior-art search, iteration, and the habit of documenting decisions as you make them. The logbook is the same instrument professional engineers and patent attorneys rely on
Recognition	Judged on a published rubric: Invention Process (40 points), Invention Impact, Inventor Communication, plus an independent originality score. Students advance from local to state affiliate to U.S. Nationals, and on to Invention Convention Globals. Patent support and cash awards are offered by many affiliates
Key dates	Affiliate competitions run through the winter and spring. U.S. Nationals: 9–11 June 2027 , at The Henry Ford in Dearborn, Michigan
Investment	Affiliate-level participation is often free through a school. Nationals costs \$365 per participant and \$390 per accompanying guest , plus travel and lodging; a parent or guardian must attend
Where a mentor fits	Mentoring is welcome throughout the build. The presentation itself must be the student's own: grades 3–4 may use notecards but receive no adult prompting of any kind; grades 5–12 use neither
Where it leads	Directly into the same competition at grades 6–8 and 9–12, then Globals — the cleanest continuous pathway in this guide
Official site	https://inhub.thehenryford.org/icw/home

2. FIRST LEGO League

The most recognized entry point into competitive robotics worldwide — and a program in transition. Teams build and program a robot while separately researching a real-world problem and pitching a solution, so the season trains engineering and communication at once.

Sector	Robotics & Engineering
Who it's for	2026–27 runs in two editions. Founders Edition: Explore, ages 6–10; Challenge, ages 9–16. Future Edition: a K–2 / ages 5–7 option and a grades 3–8 / ages 8–16 option built on LEGO Education Computer Science & AI kits
What students produce	A competition robot, the code that drives it, and an Innovation Project — original research into a community problem, presented to judges
What it builds	Iterative engineering, teamwork under a deadline, and early public presentation of technical work. The Innovation Project is most students' first experience of research that ends in a recommendation rather than a grade
Recognition	Regional and state tournaments with advancement; judged awards for Core Values, Innovation Project and Robot Design
Key dates	2026–27 season game: BIOGLOW . Team registration runs through the season; tournaments are held from late autumn into spring
Investment	Team registration, plus robot kit and tournament travel. Fees increased for 2026–27; FIRST publishes current pricing and offers team grants
Where a mentor fits	An adult coach is structural to the program — mentoring is the expected model, not an exception
Where it leads	FIRST Tech Challenge (grades 7–12) and FIRST Robotics Competition (grades 9–12), both of which continue unchanged
Worth knowing	2026–27 is the final season of FIRST LEGO League. The FIRST and LEGO Education partnership concludes after BIOGLOW, and LEGO Education launches a successor program, LEGO League, in August 2027 . Families starting now should understand they are joining a final season with a named continuation — not a program disappearing

Official site	https://www.firstinspires.org/programs/fil/ · Successor: https://education.lego.com/en-us/lego-league/program-update/
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3. Toshiba/NSTA ExploraVision

Now in its 35th year, and the rare science competition that asks students to reason forward rather than report backward. Teams pick a current technology and project it twenty years into the future, grounding the projection in real science and a literature review.

Sector	Scientific Research · Technology Forecasting
Who it's for	Grades K–12, in four bands: K–3, 4–6, 7–9, 10–12 , set by the oldest team member. Teams of two to four
What students produce	A written technology projection with a bibliography, plus a set of web-page storyboards presenting the future technology to a general audience
What it builds	Literature review, evidence-based extrapolation, and the discipline of separating what is known from what is plausible — the core move in any research proposal
Recognition	National first-place team members each receive a \$10,000 U.S. Series EE savings bond at maturity ; second place, \$5,000 each. National winners travel to a Washington, D.C. awards weekend with parents and their coach, expenses paid. Regional winners also receive awards
Key dates	Submission deadline: 27 January 2027, 8:00 PM ET
Investment	Free
Where a mentor fits	A coach is required — projects are submitted by the coach on the team's behalf, making this a natural fit for a mentored team
Where it leads	The same competition's 10–12 band, with the same artifact and the same judging standard
Official site	https://www.exploravision.org/

4. Technovation Girls

A full technology-venture season compressed into one school year, at no cost. Girls identify a problem in their own community, build a working mobile app that addresses it, and pitch the venture behind it.

Sector	Computer Science & AI · Entrepreneurship
Who it's for	Girls ages 8–18. Beginner 8–12 , Junior 13–15, Senior 16–18
What students produce	A functioning mobile application, a business plan, and a pitch video
What it builds	Mobile development, user research, and the ability to explain why a product should exist — technical and commercial fluency in the same project
Recognition	Regional judging, semifinalists and finalists, with cash awards at the global level
Key dates	The season runs from autumn to spring, with submissions in late spring. Technovation publishes the current season calendar when registration opens
Investment	Free
Where a mentor fits	Mentors are optional by rule, and the organization is explicit that teams with mentors learn more and finish more reliably. One of the least restrictive mentoring environments in this guide
Where it leads	The Junior and Senior divisions of the same program, with the same artifact type

Official site	https://www.technovation.org/
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5. VEX IQ Robotics Competition

The most accessible competitive robotics platform for elementary and middle school, and the one whose documentation requirement transfers most directly to engineering practice.

Sector	Robotics & Engineering
Who it's for	Elementary and middle school students, competing in separate divisions
What students produce	A competition robot, and a judged Engineering Notebook recording design decisions, build iterations and testing across the season
What it builds	Design iteration, and — through the notebook — technical writing under a rubric. Judges reward documentation that shows reasoning, not just outcomes
Recognition	Excellence, Design and Innovate awards alongside match performance, with advancement to the VEX Robotics World Championship
Key dates	2026–27 game: Level Up , released 30 April 2026. Events run region by region from autumn through spring
Investment	Team registration through RobotEvents, a platform fee per event registration, and event fees set by individual hosts. The robot kit and field elements are the substantial cost. Totals vary by region
Where a mentor fits	Adult coaching is standard, and the notebook rubric explicitly rewards student-authored work
Where it leads	VEX V5 Robotics Competition at middle and high school, then VEX U — same organization, same notebook discipline
Official site	https://www.vexrobotics.com/iq/competition/viqc-current-game · Events: https://events.vex.com/robot-competitions/vex-iq-competition

6. NFTE World Series of Innovation

An entrepreneurship challenge built around the UN Sustainable Development Goals, open to a wider age range than anything else in this guide.

Sector	Entrepreneurship & Business · Global Leadership
Who it's for	Ages 5–24 for the 2026–27 season; the 13–24 tier competes as the Impact League
What students produce	An innovation concept responding to a sponsor-set Impact Challenge, with a business case behind it
What it builds	Opportunity recognition, market reasoning, and the practice of aligning a venture to a defined social goal
Recognition	Cash prizes from \$300 to \$1,500 , with winners announced in spring 2027
Key dates	Season opened September 2026. Submissions due 13 December 2026
Investment	Free
Where a mentor fits	Educator and mentor support is the norm
Where it leads	Nowhere else needed — one program spans ages 5 to 24
Official site	https://innovation.nfte.com/

GRADES 6–8

7. Thermo Fisher Scientific Junior Innovators Challenge

The premier middle school science research competition in the United States, and a program a student cannot simply enter. Nomination requires placing in the top 10% of sixth through eighth graders at a Society-affiliated science fair — the same fair network that qualifies students for ISEF.

Sector	Scientific Research
Who it's for	Grades 6, 7 and 8, by nomination following competition at a Society for Science–affiliated fair in the U.S. or a U.S. territory
What students produce	A completed science fair research project, then a full online application; finalists compete in team-based engineering and science challenges
What it builds	Experimental design and data interpretation at a level judged by working scientists, plus the collaborative problem-solving the finals demand
Recognition	Highly selective: nominees narrow to a Top 300, then 30 finalists . Top award is the \$25,000 Thermo Fisher ASCEND Award ; category first and second places carry \$3,500 and \$2,500 toward educational experiences plus an iPad; every finalist receives \$500 , from a prize pool exceeding \$100,000
Key dates	Governed by the affiliated fair calendar, which runs autumn through spring; the JIC application follows nomination
Investment	No entry fee for JIC itself; affiliated fairs charge little or nothing. Finalist travel is covered
Where a mentor fits	Mentored research is expected, following the affiliated-fair convention that adult and institutional contributions are disclosed
Where it leads	The strongest continuation in this guide. The same organization runs Regeneron ISEF and the Regeneron Science Talent Search, and the affiliated-fair route a JIC nominee already knows is the ISEF qualification route
Official site	https://www.societyforscience.org/jic/

8. 3M Young Scientist Challenge

The competition whose prize is a summer working alongside a corporate research scientist. Entry is a short video; what follows for the ten finalists is a genuine mentored product development cycle.

Sector	Invention & Product Design · Scientific Research
Who it's for	Grades 5–8, entering individually
What students produce	A one-to-two-minute video describing an original invention that solves an everyday problem. The ten finalists then build a working prototype over the summer
What it builds	The ability to state a problem and a solution compellingly in two minutes, followed — for finalists — by real prototype development under professional supervision
Recognition	Ten national finalists; the winner takes \$25,000 and the title of America's Top Young Scientist. The finalist mentorship with a 3M scientist is itself among the most substantive awards available to a middle schooler
Key dates	The entry window runs from December through April each season
Investment	Free

Where a mentor fits	Each finalist is formally paired with a 3M scientist for the summer prototype assignment — mentorship is the award
Where it leads	No 3M program at 9–12, but the video-pitch-then-prototype sequence maps directly onto the Conrad Challenge and Invention Convention at high school, and finished prototypes frequently become ISEF-affiliated fair projects
Official site	https://youngscientistlab.com/annual-challenge/about-the-challenge/

9. NASA TechRise Student Challenge

Sixty student teams a year design an experiment, build it, and watch it fly. Very few competitions at any age end with the student's own hardware leaving the ground.

Sector	Aerospace & Space Science · Engineering
Who it's for	Grades 6–12. Minimum four students per team plus a team lead; all team members must attend the same U.S. public, private or charter school
What students produce	First an experiment proposal; then, for winning teams, the actual flight hardware, built into a NASA-supplied flight box
What it builds	Experimental design against real engineering constraints — mass, power, volume, and a launch date that does not move
Recognition	60 winning teams are selected. Each receives \$1,500 to build the experiment, a starter kit including the flight box, and an assigned slot on a NASA-sponsored high-altitude balloon flight
Key dates	Proposals due 2 November 2026, 11:59 PM PT. Completed experiments must reach Future Engineers by 19 May 2027 , flying in summer 2027
Investment	Free — winning teams are funded, not charged
Where a mentor fits	An educator team lead is required, and NASA technical support accompanies the build phase
Where it leads	The same program covers 9–12, and the build phase is a genuine engineering apprenticeship
Official site	https://www.futureengineers.org/nasatechrise

10. Genes in Space

The winning experiment is carried out by astronauts aboard the International Space Station. Students design real molecular biology, and finalists refine their proposals with working scientists before the final judging.

Sector	Aerospace & Space Science · Biotechnology
Who it's for	Grades 7–12, attending a U.S. public, private or home school. Individuals or teams
What students produce	A written proposal for a biotechnology experiment that could only be answered in microgravity
What it builds	Hypothesis construction and experimental design in molecular biology, and the specific skill of writing a proposal that a scientific panel will fund
Recognition	Five finalist teams are selected and paired with scientist mentors; the winning experiment flies to the ISS and is performed in orbit , with results presented at the ISS Research and Development Conference
Key dates	The annual submission deadline falls in April
Investment	Free

Where a mentor fits	Every team must have an adult sponsor; finalists receive formal scientist mentorship to develop the proposal further
Where it leads	The 7–12 band means no handoff is needed, and the proposal artifact is precisely what ISEF, JSHS and the Science Talent Search reward
Official site	https://www.genesinspace.org/

11. eCYBERMISSION

A free, fully virtual research competition run by the U.S. Army Educational Outreach Program — no travel, no equipment barrier, and a written deliverable that reads like a professional report.

Sector	Scientific Research · Engineering
Who it's for	Grades 6–9. Teams of two to four students with an adult Team Advisor
What students produce	A Mission Folder : a structured written report identifying a problem in the team's own community and either investigating it scientifically or solving it through engineering design
What it builds	The full research cycle in written form — question, method, data, conclusion — applied to a problem the students can see from their own front door
Recognition	Regional and national awards. Team Advisors can apply for Mini-Grants funding project supplies and a professional stipend
Key dates	Registration closes 24 February 2027 — advisors and students must be fully registered on complete teams by then
Investment	Free , and entirely virtual
Where a mentor fits	An adult Team Advisor is mandatory and registers the team
Where it leads	Grade 9 sits inside the eligibility band, so students carry it into high school; the Mission Folder is a direct precursor to a JSHS research paper
Official site	https://www.ecybermission.com/

12. Future City

Four deliverables, one city, and an engineer mentor built into the team structure. For the cost of a school lunch, a class gets a season-long engineering design project with a national final in Washington, D.C.

Sector	Robotics & Engineering · Urban Design
Who it's for	Grades 6–8
What students produce	Four distinct artifacts: a written essay, a physical scale model, a project plan, and a live presentation defended before judges
What it builds	Systems thinking — students must reconcile infrastructure, environment, budget and human need in a single design — plus the project management that four parallel deliverables demand
Recognition	Regional competitions in January feed a national final in Washington, D.C. Regional winners receive airfare and hotel for five: three presenting students, the educator and the mentor
Key dates	2026–27 theme: Fire Resilient Future — engineer an urban environment that prevents, withstands and recovers from wildfire. Registration closes in late October 2026 , by region; regional competitions are held in January

Investment	\$25 per organization, covering every team that organization enters , plus a \$100 cap on model materials and regional travel. The most cost-efficient serious competition in this guide
Where a mentor fits	An engineer mentor is part of the standard team structure and is one of the five people whose travel to the national final is funded
Where it leads	Future City High School , grades 9–12 — teams of three to five with an adult coach, building a digital city model in Revit alongside an essay, judged in a virtual qualification round with the top 20 advancing to Washington, D.C.
Official site	https://futurecity.org/ · High school: https://futurecity.org/future-city-high-school/

13. Congressional App Challenge

Every congressional district names a winner, and the winning apps are displayed in the U.S. Capitol. The most accessible serious computer science competition in the country.

Sector	Computer Science & AI
Who it's for	Middle and high school students; individuals or teams of up to four. Students compete in the district where they live or attend school. U.S. residents; no citizenship requirement
What students produce	An original working application in any language or platform, plus a demonstration video
What it builds	End-to-end software development — scoping, building, debugging and demonstrating something that runs
Recognition	One winner per congressional district, judged on originality, implementation and coding excellence. Winning apps are eligible for display in the Capitol Building and on House.gov, and winners are invited to #HouseOfCode on Capitol Hill each spring, which last year brought 360 students from more than 220 districts to Washington
Key dates	The 2026 Challenge ran 1 May to 26 October 2026, 12:00 PM ET
Investment	Free
Where a mentor fits	Teacher and mentor support is expected and common — the organization's own data shows most entrants work with one. AI tools are permitted provided all use is disclosed and does not constitute the entire technical development
Where it leads	One program spanning middle and high school
Official site	https://www.congressionalappchallenge.us/

14. C-SPAN StudentCam

A national documentary competition with a \$100,000 annual prize pool, now in its 23rd year. Students research a public policy question and make a film about it, with a strict runtime that forces editorial discipline.

Sector	Film, Media & Communication · Civics
Who it's for	Grades 6–12, competing in Middle School and High School categories. Individuals or teams of two or three
What students produce	A documentary of five to six minutes on the annual theme, incorporating C-SPAN footage relevant to the topic
What it builds	Research, interviewing, scripting, shooting and editing — and, through the runtime limit, the editorial judgment to cut good material that does not serve the argument. Films outside 5:00–6:00 are disqualified in the first round
Recognition	\$100,000 in cash prizes annually across the top 150 documentaries, plus awards to the teacher advisers of the top 50. Grand Prize: \$5,000 to the students and \$750 to the teacher and school

Key dates	Submissions close 20 January 2027, 11:59 PM PST
Investment	Free
Where a mentor fits	Teacher advisers are formally recognized and separately awarded
Where it leads	The same competition's High School category, with an identical artifact
Official site	https://www.studentcam.org/

15. Scholastic Art & Writing Awards

The oldest and most prestigious recognition program for creative teenagers in the United States, with alumni running from Andy Warhol and Sylvia Plath to Stephen King and Joyce Carol Oates.

Sector	Arts & Design · Writing & Humanities
Who it's for	Grades 7–12, age 13 and up, in the U.S., U.S. territories, military bases and Canada excluding Quebec
What students produce	Single works or full portfolios across 28 categories , spanning visual art, design, photography, fiction, poetry, journalism and more
What it builds	Sustained creative practice judged by working artists and writers, and — for portfolio entrants — the ability to assemble a coherent body of work
Recognition	Regional Gold and Silver Keys, then National Medals, with scholarships including Best-in-Grade and the National Student Poets Program. Regional adjudication makes this genuinely competitive at every level
Key dates	The 2027 season opened 1 October 2026 . Deadlines are regional: the earliest is 1 December 2026 , and the latest regions run to 4 January 2027
Investment	\$15 per individual entry, \$40 per portfolio entry . Fee waivers are available where cost is a barrier
Where a mentor fits	Teacher and mentor guidance is standard. Work must be the student's own and original — not copied and not AI-generated
Where it leads	The program itself runs through grade 12; a seventh-grade entrant has six years in the same competition
Official site	https://www.artandwriting.org/

16. Zero Robotics

Students write flight software; the finals run live aboard the International Space Station. Administered by MIT with NASA, using the Astrobees free-flying robots.

Sector	Aerospace & Space Science · Computer Science
Who it's for	Separate middle school and high school tournaments
What students produce	Working control software for NASA's Astrobees robots, developed and tested through MIT's online simulation environment
What it builds	Real algorithmic problem-solving under physical constraints — students debug against orbital mechanics rather than a grading rubric
Recognition	Finalist code is executed on the ISS with the competition watched live
Key dates	High school registration historically opens in October , with the tournament concluding in March; the middle school program runs in summer

Investment	Free to compete — development is done in MIT's browser-based environment
Where a mentor fits	Teams are typically coached by a teacher or an MIT-affiliated mentor
Where it leads	The middle school tournament feeds the high school tournament directly
Official site	https://zerorobotics.mit.edu/

17. The Junior Academy

Every team gets an assigned STEM mentor. The New York Academy of Sciences runs ten-week Innovation Challenges in which international student teams solve problems set by corporate and NGO sponsors.

Sector	Global Leadership & Social Impact · Scientific Research
Who it's for	Students ages 13–17. Admission to the Junior Academy is itself competitive
What students produce	A research-driven solution to a sponsor-set global challenge, developed over ten weeks with an international team
What it builds	Remote collaboration across time zones, rapid research synthesis, and working to a brief written by an external organization rather than a teacher
Recognition	Winning-team recognition, sponsor engagement, and Academy credentials
Key dates	Innovation Challenges run each autumn and spring semester , ten weeks each
Investment	Free for admitted students
Where a mentor fits	Each team is assigned a mentor — a graduate student, postdoc or working STEM professional. The Academy actively recruits mentors, with applications due in early July
Where it leads	The 13–17 band carries students from middle school into high school within one program
Official site	https://www.nyas.org/learning/high-school-research-programs/the-junior-academy/

GRADES 9–12

18. Regeneron Science Talent Search

The oldest and most prestigious science competition for high school students in the United States, running since 1942. Alumni include thirteen Nobel laureates. First place is \$250,000 — the largest science prize available to an American high school student.

Sector	Scientific Research
Who it's for	Students in their final year of secondary school in the U.S. and its territories, regardless of citizenship; U.S. citizens attending school abroad may also qualify
What students produce	An original research project, submitted with a full research report, essays, recommendations and transcript
What it builds	Independent research at a level evaluated by practising scientists, and the capacity to present it in writing to a demanding panel
Recognition	\$3.1 million awarded annually. 300 Scholars receive \$2,000 each, with a matching award to their schools; 40 Finalists receive at least \$25,000 and a week in Washington, D.C.; first place is \$250,000

Key dates	Application deadline: 5 November 2026, 8:00 PM ET. Recommendations are due at the same moment, without exception
Investment	Free
Where a mentor fits	Mentored research is the norm; Society for Science publishes dedicated guidance for the educators and mentors supporting entrants
Official site	https://www.societyforscience.org/regeneron-sts/

19. Regeneron International Science and Engineering Fair

The world's largest pre-college science competition. Roughly 1,600 finalists from more than 60 countries compete for close to \$9 million, and a student reaches it only by winning the right at an affiliated fair.

Sector	Scientific Research
Who it's for	Grades 9–12 or equivalent, qualifying through a Regeneron ISEF-affiliated fair . Team projects are capped at three members, who must be the same students who competed at the affiliated fair
What students produce	An original research project with a formal research plan, abstract, paper, poster display and a judged oral defence
What it builds	The complete apparatus of scientific practice — ethics approval, controls, statistical treatment and defence of methodology to expert judges
Recognition	Top awards include the George D. Yancopoulos Innovator Award at \$100,000 , two Regeneron Young Scientist Awards at \$75,000 each, the Gordon E. Moore Award at \$50,000 , the Leighton and Berger Family Prize at \$50,000, and four further \$10,000 awards, alongside category grand awards, special awards, scholarships and internships
Key dates	Affiliated fairs run autumn through spring. Regeneron ISEF 2027: Los Angeles, California, 8–14 May 2027. Critically, ISEF research-plan and approval paperwork must be completed before research begins
Investment	No ISEF entry fee; affiliated fairs charge little or nothing, and finalist travel is usually funded by the affiliated fair
Where a mentor fits	Research alongside adult researchers in an institution is explicitly permitted; the rules require clear acknowledgment of each individual's role and of the student's independence from the wider laboratory
Official site	https://www.societyforscience.org/isef/ · Find a fair: https://findafair.societyforscience.org/

20. Junior Science and Humanities Symposium

University-hosted, Department of Defense-sponsored, and free — students present original research to academic judges and defend it in person.

Sector	Scientific Research
Who it's for	Grades 9–12 in public, private or home schools. Students must be U.S. citizens or legal permanent residents
What students produce	An abstract and full research paper, presented orally or as a poster and defended before judges at a regional symposium hosted by a university
What it builds	The conference experience in miniature — abstract submission, oral presentation, and live defence in an academic setting, which is precisely the environment these students will enter as undergraduates
Recognition	Regional awards, with the top two delegates from each region advancing to the National Symposium to compete for military-sponsored undergraduate tuition scholarships

Key dates	Regional symposia run in late winter and spring; regional dates and deadlines are published by each host university in the autumn. The National Symposium is held in April
Investment	No cost to register or participate in a regional symposium
Where a mentor fits	Mentored research is expected. JSHS formalizes this through a required Statement of Outside Assistance submitted alongside the paper — disclosure, not restriction
Official site	https://www.jshs.org/ · Find your region: https://www.jshs.org/students/find-your-region/

21. Davidson Fellows Scholarship

Scholarships of \$100,000, \$50,000 and \$25,000 for work at or near the college-graduate level — the highest individual award amounts available to a pre-college student for a single body of work.

Sector	Scientific Research · Writing & Humanities · Arts
Who it's for	Age 18 or younger at the deadline, with no minimum age. U.S. citizens or Permanent Residents residing in the U.S., or students stationed overseas on active U.S. military duty
What students produce	A completed significant piece of work in Science, Technology, Engineering, Mathematics, Literature, Music, Philosophy or Outside the Box
What it builds	This one rewards rather than builds — it is the natural destination for a student whose multi-year project has outgrown the usual competitions
Recognition	\$100,000 / \$50,000 / \$25,000 scholarships. The stated bar is work at or close to college-graduate level, with depth of knowledge in the field and genuine potential to benefit society
Key dates	The application opens in the autumn , with the deadline in February
Investment	Free
Where a mentor fits	Mentored work is common among Fellows; the work must be demonstrably the student's own creation
Official site	https://www.davidsongifted.org/gifted-programs/fellows-scholarship/

22. MIT THINK Scholars Program

The only competition in this guide that rewards a student for having a project they cannot yet execute alone. THINK exists for students who have done the background research but lack the resources and guidance to build the thing.

Sector	Scientific Research · Engineering
Who it's for	High school students residing in the United States
What students produce	A written research proposal — not a completed project — for a novel science, technology or engineering idea, including an explicit case for how the project would benefit from funding and mentorship
What it builds	Proposal writing, which is the central professional skill in research and the one least often taught before graduate school
Recognition	Finalists receive mentorship from MIT students, professors and researchers in their field, project funding, and a visit to MIT
Key dates	The application cycle runs in the autumn with a deadline around the turn of the year; MIT THINK publishes program guidelines for each cycle
Investment	Free

Where a mentor fits	Mentorship is the prize. No other program on this list is so precisely designed around the handoff from a student's ambition to an expert's guidance
Official site	https://think.mit.edu/

23. Stockholm Junior Water Prize

The world's most prestigious youth water science award, with the U.S. national winner advancing to the international final in Stockholm.

Sector	Environment & Sustainability · Scientific Research
Who it's for	Grades 9–12, having reached age 15 by 1 August of the competition year. Teams of up to two. Residents of the 50 states, D.C. and Puerto Rico
What students produce	An original water-related research project with a full written research paper
What it builds	Applied environmental research with a clear real-world stake, and the discipline of writing for a panel of water professionals
Recognition	State competitions feed a national final; state prizes vary and are published state by state. The U.S. winner represents the country at the international Stockholm Junior Water Prize
Key dates	State entry deadlines fall in April , with state winners judged in early May and national finals in June
Investment	Free to enter at state level; travel to the national final
Where a mentor fits	Mentored laboratory and field research is standard, following the same disclosure conventions as the affiliated-fair network
Official site	https://www.wef.org/sjwp/

24. Lemelson-MIT InvenTeams

A grant, not a contest. Teams of high school students, a teacher and mentors receive up to \$7,500 and a year of MIT support to invent a technological solution to a problem they chose themselves.

Sector	Invention & Product Design · Engineering
Who it's for	High school teams — students together with a teacher and mentors
What students produce	A working technological invention, developed across a full academic year
What it builds	Everything a funded project demands and an unfunded one does not: budgeting, procurement, milestone reporting, and delivering to a grantor
Recognition	Grants of up to \$7,500 per team , plus a year of structured support from the Lemelson-MIT Program
Key dates	The initial application is posted each October and due in early April , for grants awarded for the following academic year. Selected teams submit a final application in September
Investment	Free to apply — the award is funding
Where a mentor fits	Mentors are named in the funded unit itself. Of everything in this guide, this is the program that most explicitly treats the mentor as part of the team
Official site	https://lemelson.mit.edu/inventeams

25. World Food Prize Youth Institutes

Free, mentor-supervised, and the top outcome is a paid international research internship. Students write on global food security and defend their thinking at a state or virtual institute, with the strongest advancing to the Global Youth Institute.

Sector	Agriculture & Food Security · Global Leadership
Who it's for	Grades 9–12 across most state institutes; the Iowa Youth Institute accepts grades 8–12
What students produce	A three-to-five-page research paper on a chosen country and a food-security topic, written under the supervision of a teacher or mentor, then presented and discussed with experts
What it builds	Policy-grade research writing and the ability to hold a substantive conversation with professionals working in the field
Recognition	Advancement to the Global Youth Institute in Des Moines, and eligibility for the Borlaug-Ruan International Internship — an eight-week, all-expenses-paid research placement at an international agricultural research centre — plus USDA and Wallace-Carver internships
Key dates	Each state and virtual institute sets its own paper and registration deadline, generally in the spring; the Global Youth Institute is held in October
Investment	Free. No registration cost, and meals are provided at the institute. Each high school in a participating state is guaranteed two Global Youth Institute places
Where a mentor fits	The paper is written under teacher or mentor supervision by design
Official site	https://www.worldfoodprize.org/en/youth_programs/global_youth_institute/

26. Conrad Challenge

A four-phase innovation and entrepreneurship competition run by Space Center Houston, taking teams from a blank page to a live pitch in front of industry judges over a full academic year.

Sector	Entrepreneurship & Business · Invention & Product Design
Who it's for	Students ages 13–18, in teams of two to five, competing from anywhere in the world. Teams must have a coach aged 18 or over
What students produce	A Lean Canvas, then a full innovation package covering technical development, market analysis and business model, and finally a live Power Pitch at the Innovation Summit
What it builds	The complete venture-creation sequence, taught in the order founders actually encounter it, and judged by people who fund companies
Recognition	Pete Conrad Scout and Innovator awards, with seed funding and in-kind support for winning teams. Five categories span aerospace, energy, health, cyber-technology and transformation
Key dates	The 2026–27 season runs on published deadlines of 29 October 2026 , 2 November 2026 , 3 December 2026 and 7 January 2027 , with Innovation Summit registration due 25 March 2027 and Power Pitch materials due 1 April 2027
Investment	The Activation and Lean Canvas phases are free , so a team can test its idea at no cost. Advancing to the Innovation Stage carries an entry fee, and Innovation Summit attendance is \$499 per guest plus travel. Financial aid, group discounts and fee waivers are available , and teams receive instructions at each advancement
Where a mentor fits	Every team must have an adult coach as supervisor; mentoring through the technical and business development phases is expected
Official site	https://conrad.spacecenter.org/

27. Diamond Challenge

A free, university-hosted venture competition with a **\$100,000 prize pool**, run by Horn Entrepreneurship at the University of Delaware, with separate tracks for commercial and social ventures.

Sector	Entrepreneurship & Business · Social Innovation
Who it's for	Teams of two to four high school students aged 14–18 at the submission deadline
What students produce	A written concept narrative and introductory video in round one, developing into a full venture and live pitch in later rounds
What it builds	Customer discovery, business model design and investor-facing communication — with the social track allowing students whose motivation is impact rather than profit to compete on their own terms
Recognition	\$100,000 prize pool , with awards to the top three teams in each of the Business Innovation and Social Innovation tracks, culminating at the Limitless World Summit
Key dates	Teams develop concepts from September ; three rounds of judging run February through April
Investment	Free to enter. Travel to the summit for finalists
Where a mentor fits	Adult advisers are standard throughout
Official site	https://diamondchallenge.org/

28. Blue Ocean Student Entrepreneur Competition

Free, international, and built entirely around a five-minute pitch — the lowest barrier to entry of any serious entrepreneurship competition in this guide.

Sector	Entrepreneurship & Business
Who it's for	Students enrolled in high school at registration, individually or in teams of up to five, worldwide. Students who graduate before the deadline may still submit
What students produce	A video pitch under five minutes, in English, presenting an original business concept built on blue-ocean strategy
What it builds	The ability to make a business case compellingly and briefly — the single most transferable skill in entrepreneurship
Recognition	Cash prizes to top teams; the Top 10 receive a Certificate of Outstanding Achievement, and three Blue Ocean Teacher Champions receive \$500 each
Key dates	Submissions close 21 February 2027
Investment	Free , with a Blue Ocean Mini-Course provided to help teams prepare
Where a mentor fits	Teacher and mentor involvement is recognized formally through the Teacher Champion awards
Official site	https://blueoceancompetition.org/

29. Rise

A lifetime programme, not a prize. Rise — an initiative of Schmidt Futures and the Rhodes Trust, and the anchor programme of a \$1 billion commitment by Eric and Wendy Schmidt — selects 100 Global Winners a year and supports them for life.

Sector	Global Leadership & Social Impact
Who it's for	Young people ages 15–17 worldwide, with age determined as of 1 July in the application year
What students produce	A three-stage application: video introductions, an individual project that demonstrates the applicant's talents and benefits their community , and peer review of other applicants' projects
What it builds	Self-directed service — students must find a problem in their own community, act on it, and show the result. There is no set brief, which makes it the most genuinely original piece of work most applicants will have produced
Recognition	Deeply selective: up to 500 Finalists advance to Finalist Days, an innovative interview format assessing motivation, problem-solving and teamwork, from which 100 Global Winners are chosen. Winners receive need-based scholarships, mentorship, networking, career development access and the potential for ongoing funding , extended across their working lives
Key dates	Application cycles have historically opened in the autumn and closed in mid-January . Rise publishes each cycle's dates on its own site
Investment	Free . Applications are submitted through the Hello World Network app, with browser and paper pathways for applicants without mobile access — the accessibility is deliberate and unusual
Where a mentor fits	The project is the applicant's own; mentorship in shaping and executing community work is entirely compatible with the format
Official site	https://www.risefortheworld.org/ · Rhodes Trust: https://www.rhodeshouse.ox.ac.uk/about/rise/

30. FIRST Robotics Competition & FIRST Tech Challenge

The largest robotics competition ecosystem in the world, and the one with the deepest scholarship network attached to it. Hundreds of universities and corporations reserve scholarships specifically for FIRST participants.

Sector	Robotics & Engineering
Who it's for	FIRST Tech Challenge: ages 12–18, grades 7–12. FIRST Robotics Competition: ages 14–18, grades 9–12
What students produce	A competition robot designed, built, programmed and iterated across a season, alongside a judged Engineering Portfolio documenting the process
What it builds	Engineering under genuine constraint — fixed deadlines, finite budget, a functioning machine at the end — plus the fundraising, outreach and project management a team must run alongside the build
Recognition	District, regional and championship advancement, with judged awards including Inspire, Think, Connect, Innovate and Control. The associated scholarship network is the largest attached to any competition in this guide
Key dates	2026–27 seasons are underway: FIRST Tech Challenge BIOBUZZ presented by RTX kicked off 12 September 2026 , and FIRST Robotics Competition BIOCORE runs the same season. Registration payment for opt-out options is due 17 November 2026
Investment	The largest commitment in this guide. Registration fees rose for 2026–27; FRC season registration includes event entry, and teams additionally fund robot parts, tools and travel. FIRST publishes current pricing and runs a substantial team grant programme
Where a mentor fits	Adult mentorship is structural and culturally central — FIRST teams are built around practising engineers working alongside students
Official site	https://www.firstinspires.org/programs/ftc/ · https://www.firstinspires.org/programs/frc/

31. National History Day

A year-long original research project in history, with five ways to present it, reaching more than half a million students annually through a structured affiliate network.

Sector	Writing & Humanities
Who it's for	Junior Division, grades 6–8 and Senior Division, grades 9–12
What students produce	Original historical research on the annual theme, presented as a paper, exhibit, documentary, performance or website , with a process paper and annotated bibliography in every category
What it builds	Primary-source research, historiographical argument, and the discipline of an annotated bibliography — the closest thing at high school level to genuine humanities scholarship
Recognition	Affiliate contests feed the national contest, with category awards and numerous special prizes
Key dates	2027 theme: Innovation in History: Impact, Influence, Change . Affiliate contests run winter through spring, with deadlines set by each affiliate. National Contest: 13–17 June 2027 , University of Maryland, College Park
Investment	Affiliate-dependent; national participation carries travel costs
Where a mentor fits	Teacher-supervised research is the standard model; the work must be the student's own
Official site	https://nhd.org/

32. John F. Kennedy Profile in Courage Essay Contest

The only writing competition in this guide that requires a named adult to read and improve the student's draft before submission. Mentorship is written into the rules.

Sector	Writing & Humanities · Civics
Who it's for	U.S. high school students in grades 9–12, including home-schooled students and those under 20 in correspondence or GED programmes, across the 50 states, D.C., the territories, and U.S. citizens attending school overseas
What students produce	An original research essay of 700 to 1,000 words on an act of political courage by a U.S. elected official, with full citations and bibliography excluded from the word count
What it builds	Argument under a severe word limit, supported by primary research — the hardest and most useful constraint in expository writing
Recognition	First place \$10,000 ; second place \$3,000 ; three finalists \$1,000 each; fifteen honourable mentions. Top winners receive a hardcover copy of <i>Profiles in Courage</i>
Key dates	The contest opens 1 September , with submissions due in the winter
Investment	Free
Where a mentor fits	Students must name a nominating teacher on the registration form. That teacher's defined role is to advise the student during writing and to read the essay and suggest improvements before submission
Official site	https://www.jfklibrary.org/learn/education/profile-in-courage-essay-contest

33. The Concord Review

The only quarterly journal in the world that publishes academic history papers by secondary school students — and the credential carries weight precisely because acceptance is rare.

Sector	Writing & Humanities
Who it's for	Secondary school students worldwide
What students produce	A long-form original history research paper, typically several thousand words with full scholarly apparatus
What it builds	Sustained historical argument at genuine length, with the citation discipline of academic publishing
Recognition	Publication itself is the award. Acceptance is highly selective, and published authors appear in a journal that university history faculty read
Key dates	Rolling quarterly deadlines: 1 February for the Summer issue, 1 May for Autumn, 1 August for Winter and 1 November for Spring. A submitted essay is considered for the next three issues, and issues appear in September, December, March and June
Investment	A submission fee applies — recently in the region of \$70 , which includes a year's digital subscription to the journal. Families should understand this as a non-refundable submission cost against a low acceptance rate
Where a mentor fits	Teacher-supervised research is the norm, and most published papers begin as a school assignment taken considerably further
Official site	https://tcr.org/

34. YoungArts

The most selective arts award available to American teenagers, and the sole nominating body for the U.S. Presidential Scholars in the Arts.

Sector	Arts & Design
Who it's for	Artists in grades 10–12 or ages 15–18 as of 1 December 2026
What students produce	A discipline-specific portfolio — video, audio, documents or images depending on the art form — adjudicated by panels of practising artists
What it builds	Portfolio construction and the experience of being assessed by professionals in the field against a national standard
Recognition	Two tiers. The YoungArts Award carries \$250 . The YoungArts Award With Distinction brings an invitation to National YoungArts Week and cash awards of up to \$10,000 . Winners enter a lifelong artist-support network
Key dates	Application deadline: 6 October 2026, 8:00 PM ET for the 2027 competition
Investment	An application fee applies, with fee-waiver documentation accepted
Where a mentor fits	Preparation with a teacher or private instructor is standard practice in every discipline
Official site	https://youngarts.org/apply/

35. Breakthrough Junior Challenge

A **\$250,000 college scholarship for a two-minute video** — the largest single prize in this guide, awarded for explaining one scientific idea extraordinarily well.

Sector	Film, Media & Communication · Scientific Research
Who it's for	Individuals ages 13–18 worldwide
What students produce	An original video of up to two minutes explaining a concept or theory in science or mathematics engagingly and imaginatively

What it builds	Science communication — the ability to make a hard idea land with a general audience, which is increasingly what separates researchers who get funded from those who do not
Recognition	\$250,000 college scholarship to the winner, up to \$100,000 toward a science lab at the winner's school, and \$50,000 to a teacher the winner names as an inspiration
Key dates	The competition typically opens in May and closes in mid-September
Investment	Free
Where a mentor fits	Entry is individual, and mentoring on scripting, scientific accuracy and production is entirely compatible with that
Official site	https://breakthroughjuniorchallenge.org/

36. All American High School Film Festival

The largest student film festival in the world, held annually in New York City, with official selections screened in Times Square theatres.

Sector	Film, Media & Communication
Who it's for	Filmmakers who were in high school or younger when the film was produced. Middle school students are eligible, entries are accepted internationally, and recently graduated seniors may submit before the final deadline
What students produce	A completed short film in any of numerous categories; students may submit as many films in as many categories as they wish
What it builds	Complete film production, and the experience of a festival circuit — submission, selection, screening and industry exposure
Recognition	Official Selection status, category awards and screenings in New York City, with industry panels and college recruiting attached
Key dates	The submission window opens in November , with an early free submission deadline in March and a final deadline in early July
Investment	Free if submitted by the early deadline ; entry fees apply afterward. The separate NYC Film Invitational costs \$250 for teams of five or fewer and \$500 for larger teams. Attending the festival means travel to New York
Where a mentor fits	Film teachers and mentors are standard throughout the production process
Official site	https://www.hsfilmfest.com/

37. Modeling the Future Challenge

A research competition for the quantitative student who is not in a laboratory. Teams model real-world data, quantify risk and make recommendations to an actual organization.

Sector	Data Science & Quantitative Analysis
Who it's for	U.S. high school students taking junior- or senior-level mathematics — statistics, probability, pre-calculus, calculus or similar. Submissions from one to five students
What students produce	An original research project: a mathematical model built on real data, a risk analysis, and a written report making recommendations to a company, industry group, government or organization
What it builds	Applied statistical modelling and, unusually for a high school competition, the translation of quantitative findings into a recommendation a decision-maker could act on

Recognition	\$60,000 in scholarship prizes. Finalists attend the Modeling the Future Symposium and meet practising actuaries
Key dates	Registration opens each autumn , with the season running into the spring
Investment	Free registration
Where a mentor fits	Coaches and actuarial professional mentors participate throughout
Official site	https://www.mtfchallenge.org/

38. Bow Seat Ocean Awareness Contest

An international environmental competition that accepts work in any creative medium — and requires every entry to be paired with a written reflection grounded in research.

Sector	Environment & Sustainability · Arts & Design
Who it's for	Students ages 11–18 worldwide in middle or high school. Junior Division 11–14, Senior Division 15–18
What students produce	A creative work in Visual Art (handcrafted or digital), Creative Writing, Film, Interactive & Multimedia, Performing Arts or Poetry & Spoken Word, submitted alongside a written reflection
What it builds	Environmental literacy expressed through a student's existing creative strength — the rare competition that lets an artist and a scientist compete on the same question
Recognition	Scholarship awards up to \$1,500 , plus the We All Rise Prize : ten \$500 awards for U.S. students who identify as Black, Indigenous and/or Latine
Key dates	The annual theme is announced in September , with submissions due in early June
Investment	Free
Where a mentor fits	Teacher and mentor guidance is encouraged
Official site	https://bowseat.org/programs/ocean-awareness-contest/

39. GENIUS Olympiad

An international environmental project competition that takes science, engineering, business, robotics and the arts on equal terms — hosted on a university campus in Rochester, New York.

Sector	Environment & Sustainability
Who it's for	Students in grades 8–12, worldwide
What students produce	An environment-themed project in one of eight categories: Science, Engineering, Art, Short Film, Music, Creative Writing, Robotics or Business
What it builds	Breadth — a student whose environmental interest is expressed through film or business competes in the same arena as one running experiments, which few competitions allow
Recognition	Gold, silver and bronze medals by category, with scholarship offers from partner universities. Finalists present in person at St. John Fisher University
Key dates	Applications open in December 2026 , with project submission in the spring and finals held in June in Rochester, New York

Investment	An application fee of approximately \$50 per project , and finalists pay a participation fee — recently in the range of \$425 to \$850 depending on accommodation — covering registration, campus stay, dining and excursions, plus travel
Where a mentor fits	Adult advisers are standard
Official site	https://geniusolympiad.org/

Choosing well

Three patterns are worth carrying into any conversation with a family.

Registration and submission are rarely the same date, and registration is the one students miss. The Congressional App Challenge opens in May and closes in October. NASA TechRise takes proposals in November and flight hardware the following May. The entire ISEF pathway requires approval paperwork completed *before* research begins — months ahead of any fair. A student who finds a competition in March has already missed most of the year's best ones.

The strongest mentoring fits are explicit about it. MIT THINK awards mentorship as the prize. Lemelson-MIT names mentors in the funded unit. The Junior Academy assigns one. The JFK essay contest requires a named teacher to read the draft. Future City funds the mentor's travel to the national final. These are the programmes built for the way BlueRobins works.

Cost is rarely the headline number. Future City costs \$25 per organization. eCYBERMISSION, JSHS, the World Food Prize and the Science Talent Search are free. The expensive commitments — FIRST, Conrad, Invention Convention Nationals, GENIUS Olympiad — mostly charge as a student advances, which is exactly when a family should already know it is coming.

Competition details reflect the 2026–27 season as published by each organizing body. Deadlines, fees and eligibility are set by the organizers and should be confirmed on the official site before a student commits to a season.